

AI, Photography, and Art



CS180: Intro to Computer Vision and Comp. Photo

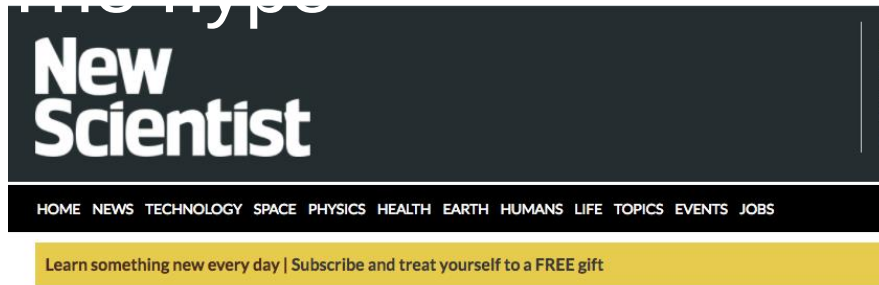
*With many slides from Aaron Hertzmann,
Yan Ke and Fredo Durand, Tamara Berg*

Alexei Efros, UC Berkeley, Fall 2025

Outline

1. Is AI killing art?
2. How to take Good Photos?
3. How not to be scared at a Modern Art museum

The Hype



Home | News | Technology

DAILY NEWS 29 June 2017

Artificially intelligent painters invent new styles of art



Art for the AI generation
Art and Artificial Intelligence Laboratory, Rutgers University

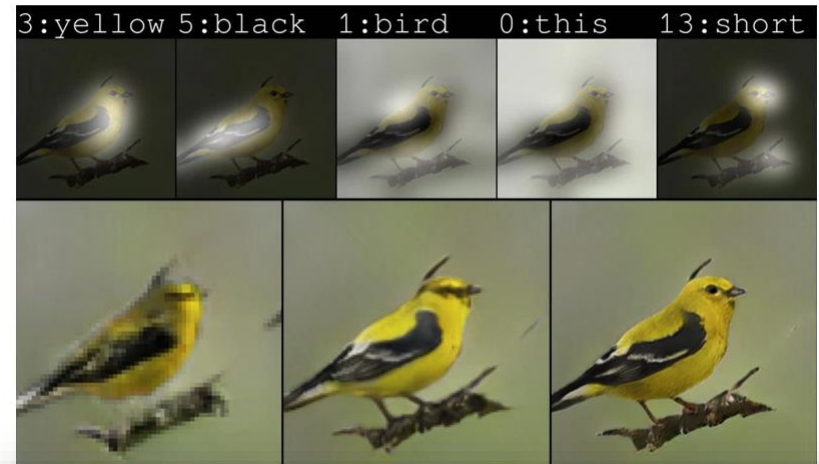
by Chris Bernini

Slide by Aaron Hertzmann

Microsoft's new drawing bot is an AI artist

Sarah Perez @sarahintampa / Jan 18, 2018

Comment



This Artificially Intelligent Robot Composes and Performs Its Own Music



George Dvorsky

6/14/17 11:25am • Filed to: WE ARE THE ROBOTS ▾

24 5



We've been here before...



Daguerre

HISTORIQUE ET DESCRIPTION

DES PROCÉDÉS

DU

DAQUERRÉOTYPE

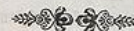
et du Diorama,

PAR DAGUERRE,

Peintre, inventeur du Diorama, Officier de la Légion-d'Honneur,
membre de plusieurs Académies, etc.

Nouvelle Edition,

CORRIGÉE, ET AUGMENTÉE DU PORTRAIT DE L'AUTEUR.



Paris,

ALPHONSE GIROUX ET C^{ie}, ÉDITEURS,

RUE DU COQ-SAINT-HONORÉ, N° 7,

où se fabriquent les Appareils;

ET CHEZ LES PRINCIPAUX LIBRAIRES, PAPETIERS,

Marchands d'Estampes et Opticiens.

1839.



“From today,
painting is
dead”

— Paul Delaroche,
1839



“This is the end of Art.
I am glad I have had my
day.”

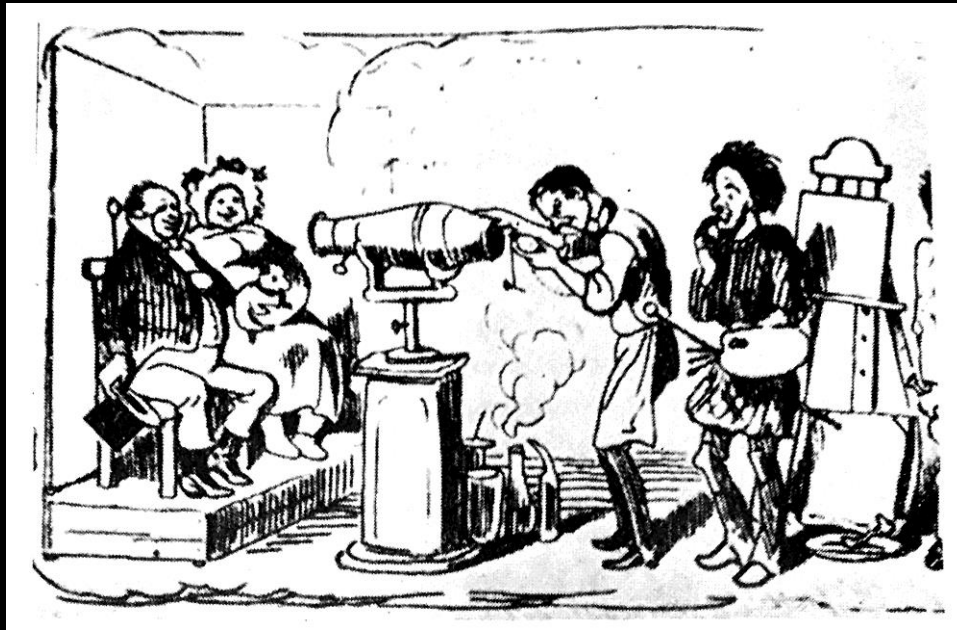
J.M.W. Turner, 1839

Portraiture in 1838





1851: Jabez Hogg Making a Portrait in Robert Beard's Studio



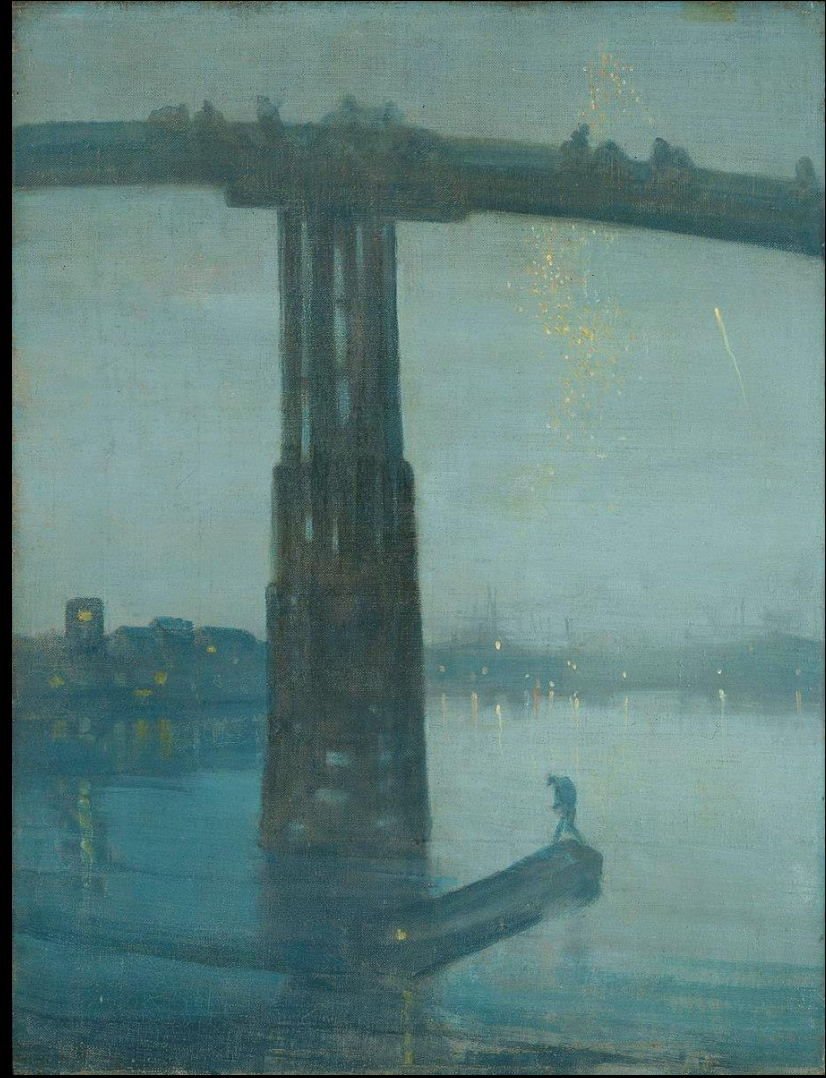
Hosemann, 1843

“One knows that *photography has harmed painting considerably, and has killed portraiture* especially, once the livelihood of the artist.”

- Henry Le Secq, painter-photographer,
1863

“The imitator is a poor kind of creature. If the man who paints only the surface he sees before him were an artist, the king of artists would be the photographer. It is for the artist to do something beyond this...”

– J.M. Whistler, 1878 (paraphrased)





“Accurate drawing is not the essential thing to aim at, because the reflection of reality in a mirror, would not be a picture at all, no more than a photograph.”

– Vincent Van Gogh, 1888 (paraphrased)

Photography: Summary

At first, it looked like photography automated art

Many artists feared and disparaged it

Actual outcomes:

These trends often repeat with each new art technology

- New art form (photography)
- Invigorated old art forms (painting)
- Jobs affected: portraiturists replaced with photo studios
- Image creation made available to hobbyists

“In those early days, animators were frightened of the computer. They felt it was going to take their jobs away. We spent a lot of time telling people, ‘No, it’s just a tool—it doesn’t do the creativity!’”

— Alvy Ray Smith, on the very early days at NYIT and Lucasfilm Graphics Group



Read Aaron Hertzmann to learn more...



Aaron Hertzmann's blog

cinema.mosfilm.ru

When Machines Change Art

At a few times in history, new technologies came along that changed the way we make art. Machines, chemicals, and/or algorithms replaced some of the steps that artists did, changing how we made art—and, sometimes, radically transforming what we thought think art is.

Now seems like one of those times.

When change happens, it's surprising, and it's tempting to see it in extremes: the new technology is either going to improve everything or ruin everything. But the reality of how past technologies changed art is nuanced and complex, with different effects for different communities in the short-range and then the long-range.

In this post, I'm specifically focused on technologies that automate (or seem to automate) steps of the artistic process. What can past examples of such technologies changing art have in common, and do they have lessons for the current moment?

https://aaronhertzmann.com/2022/12/17/when-tech-changes-art.html?fbclid=IwY2xjawG7NcJleHRuA2FlbQIxMAABHVCMzPFKsjNC7Z3GbqbJx94tnIPuOkaAolQE4BvOSK6LkrUFTIdGRHsoaw_aem_8WfQsdc4DQuzLcQ9o64U-A

What Makes a Great Picture?



© Robert Doisneau, 1955

Framing

“Photography is all about framing. We see a subject -- and we put a frame around it. Essentially, that is photography when all is said and done.”

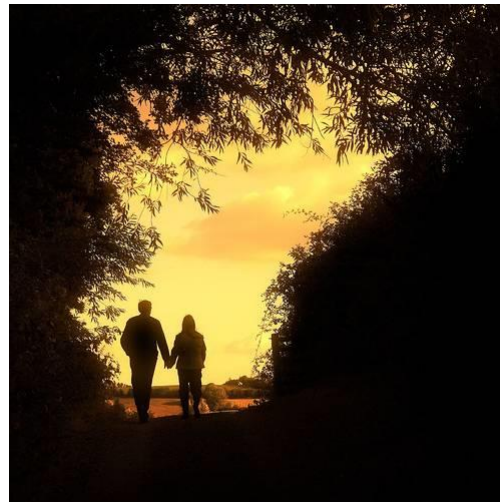
-- from photo.blorge.com



Frame serves several purposes:

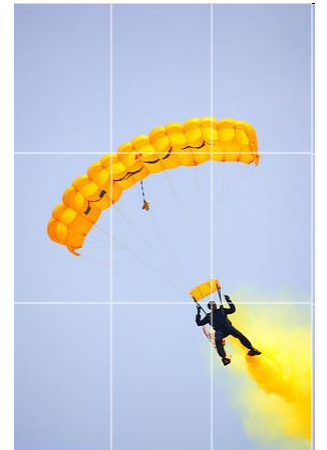
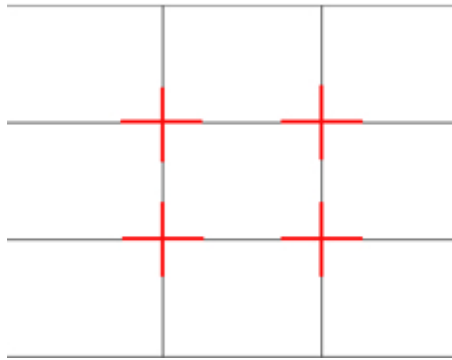
1. It gives the image depth
2. If used correctly, framing can draw the eye of the viewer of an interest to a particular part of the scene.
3. Framing can bring a sense of organization or containment to an image.
4. Framing can add context to a shot.

Examples of nice framing



<http://flickr.com/photos/paulosacramento/226545698/>
<http://flickr.com/photos/chrisbeach/13868545/>
<http://flickr.com/photos/74531485@N00/929270814/>
<http://flickr.com/photos/freakdog/223117229/>
<http://flickr.com/photos/cdm/253805482/>

Rules of Thirds



Other examples



Don't center, especially for motion



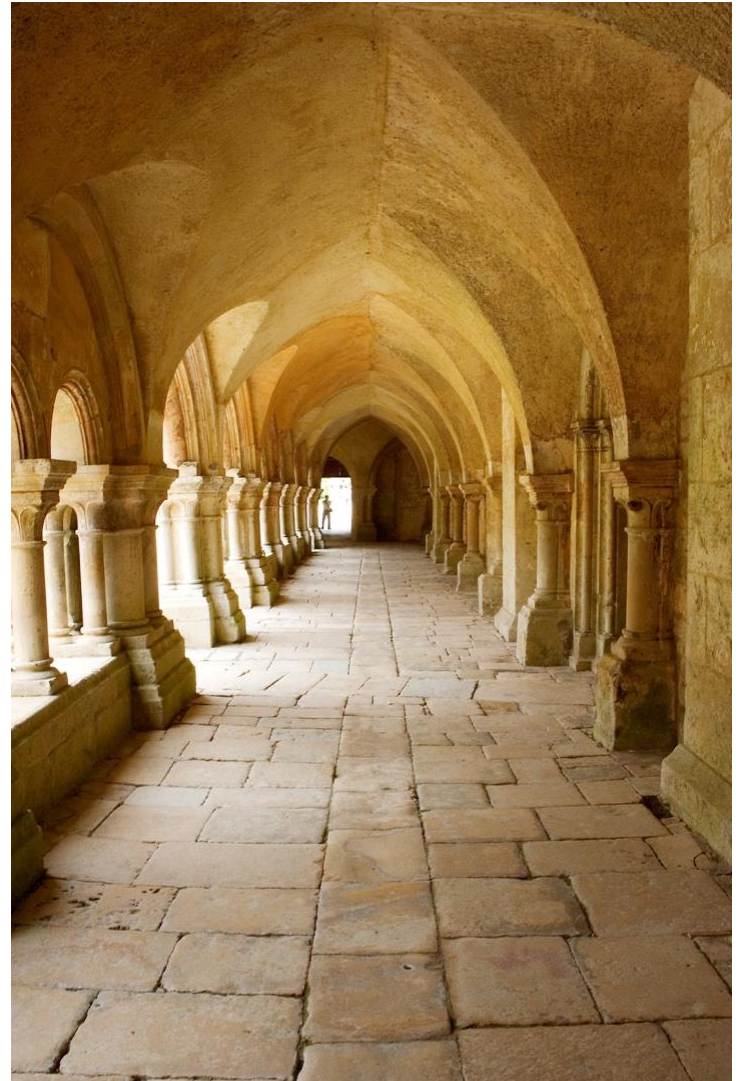
By Fredo Durand

Don't center, especially for motion



By Fredo Durand

... or do center



By Fredo Durand

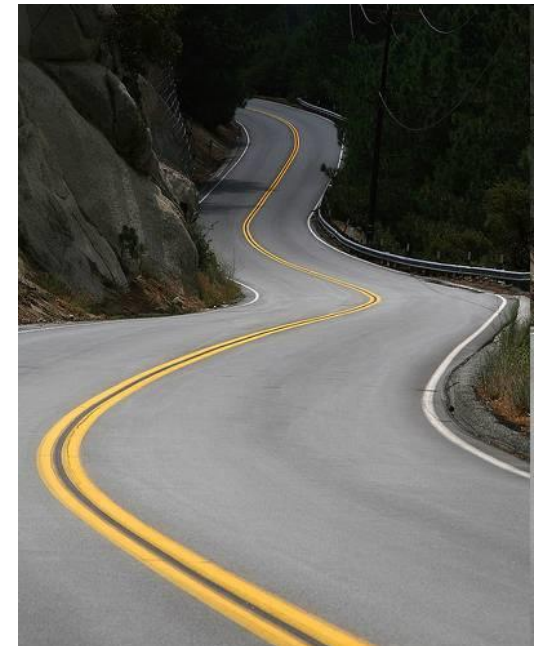
Leading Lines





By Fredo Durand

More examples



Textures and Patterns



Simplicity

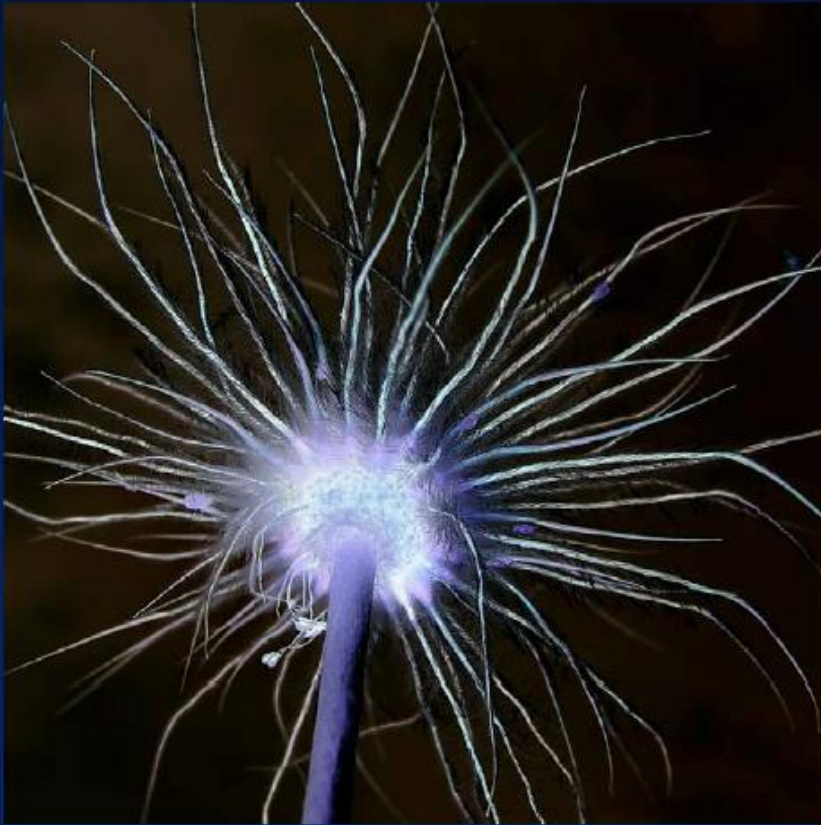


“Look Into” by Josh Brown @ Flickr



Prof - Obvious what one should be looking at
ie easy to separate subject from the
background. Snap – unstructured, busy, filled
with clutter.

Simplicity



Simplicity



B&W for Simplicity



B&W for Simplicity



...but not always



...but not always

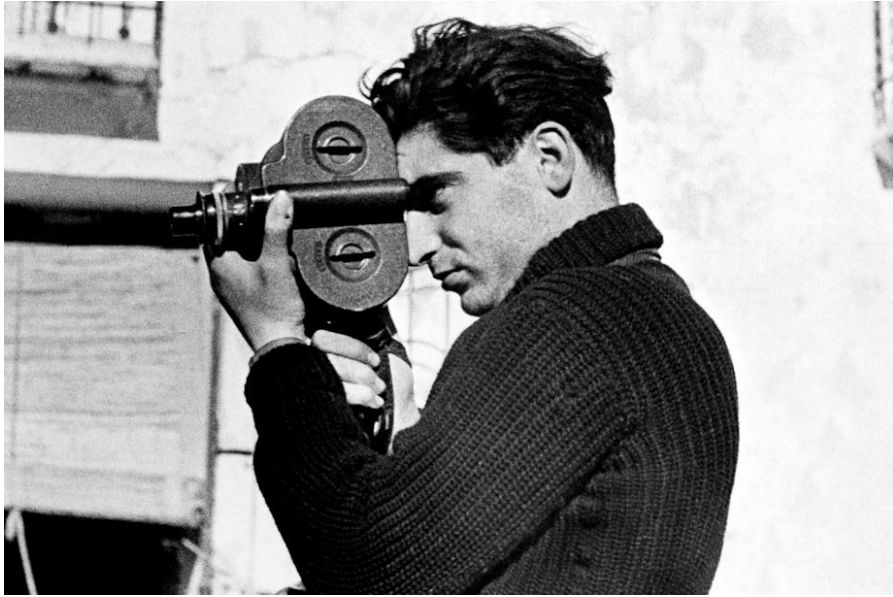


Crop for Simplicity



By Fredo Durand

If your pictures aren't good enough, you're not close enough" — Robert Capa



Simplicity for Portraits – just move



<https://vimeo.com/29722267>

Get low

Try to be at eye level



Bad



Better

By Fredo Durand



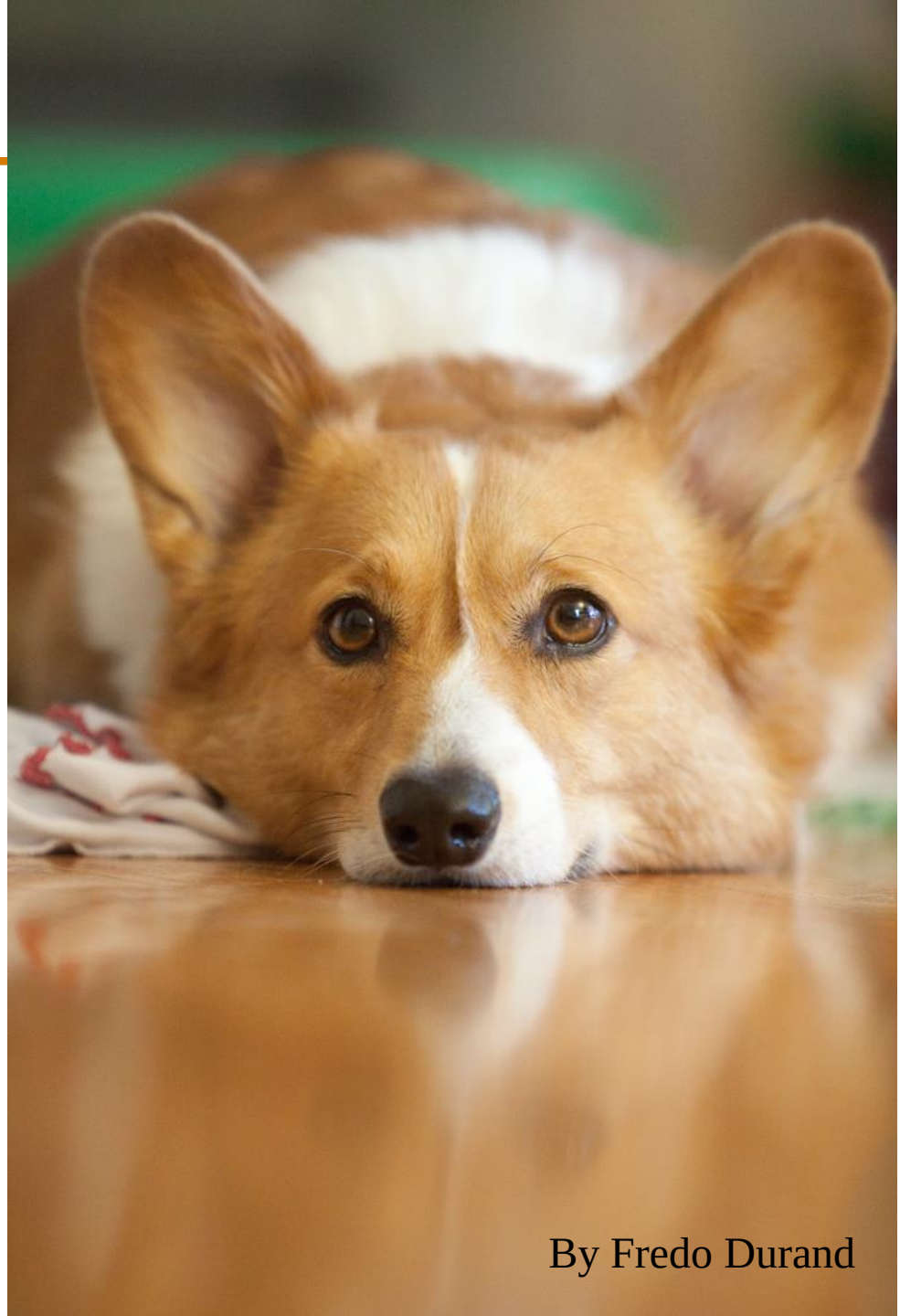
Bad angles

<https://www.youtube.com/watch?v=8EmRZO9fwvk&feature=youtu.be>



By Fredo Durand

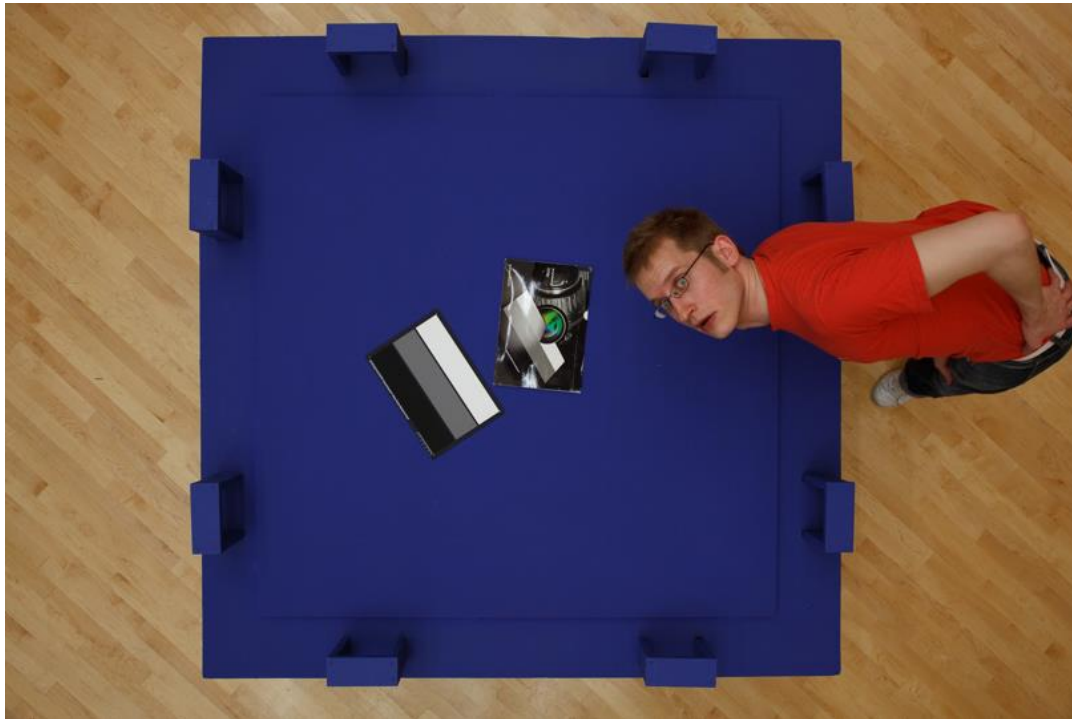
Eye level



By Fredo Durand

Or really get high

**As usual, follow a rule
or really break it.**



By Fredo Durand

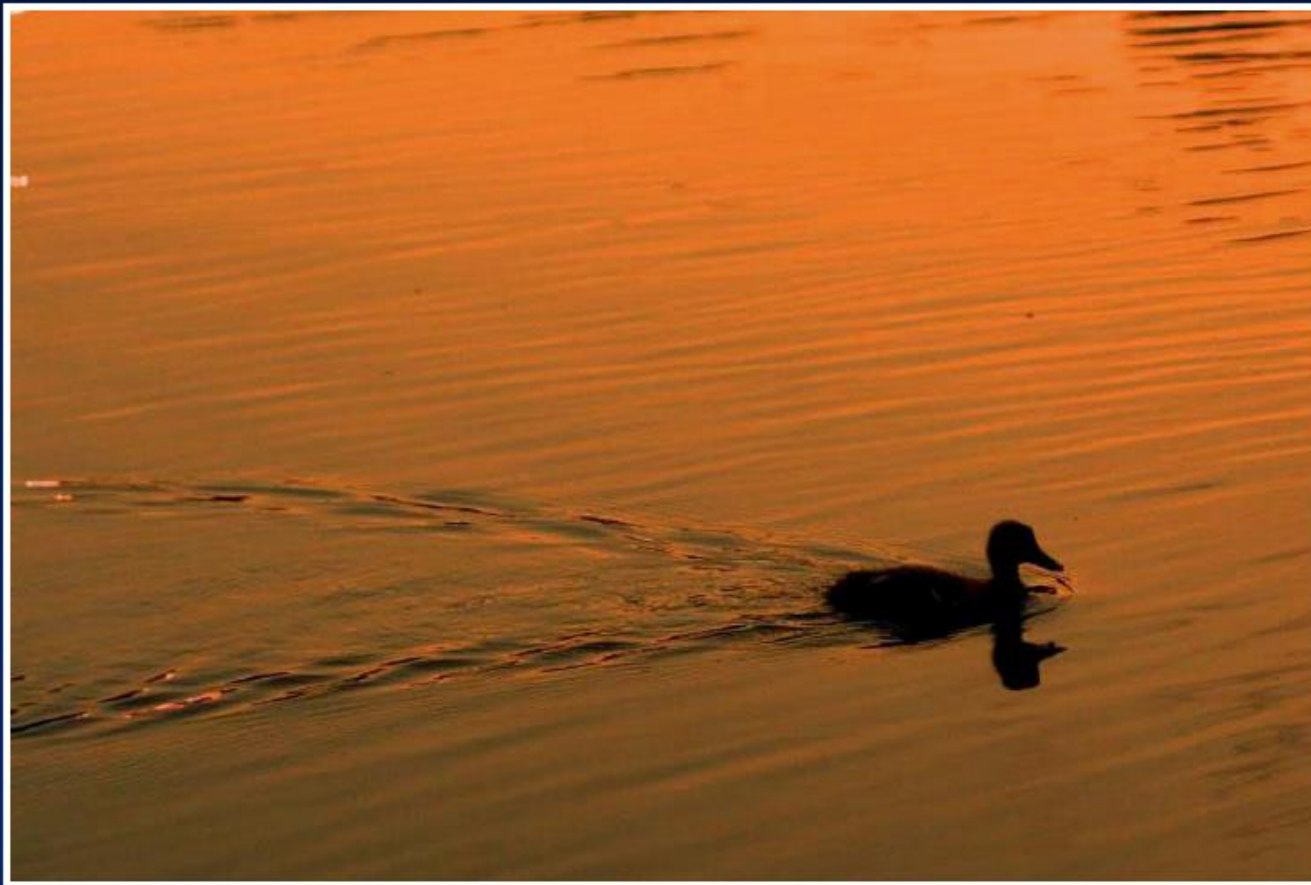
Front Lighting



Side Lighting



Back Lighting



Go in the shade

Light is more diffuse

Bad



Better



Overcast days are the best

Just don't put the sky in the frame

The weather conditions



The pictures



Other overcast-day pictures



By Fredo Durand

Bottom line

**Don't get married
on a sunny day!**



Cloudy day



Best time of day: sunset & sunrise

+/- 1 hour

“Golden hours”

Night photography: always near sunset/sunrise

- because of nice diffuse light

Mid day:
often not great

less than 1 hour
after sunrise/
before sunset

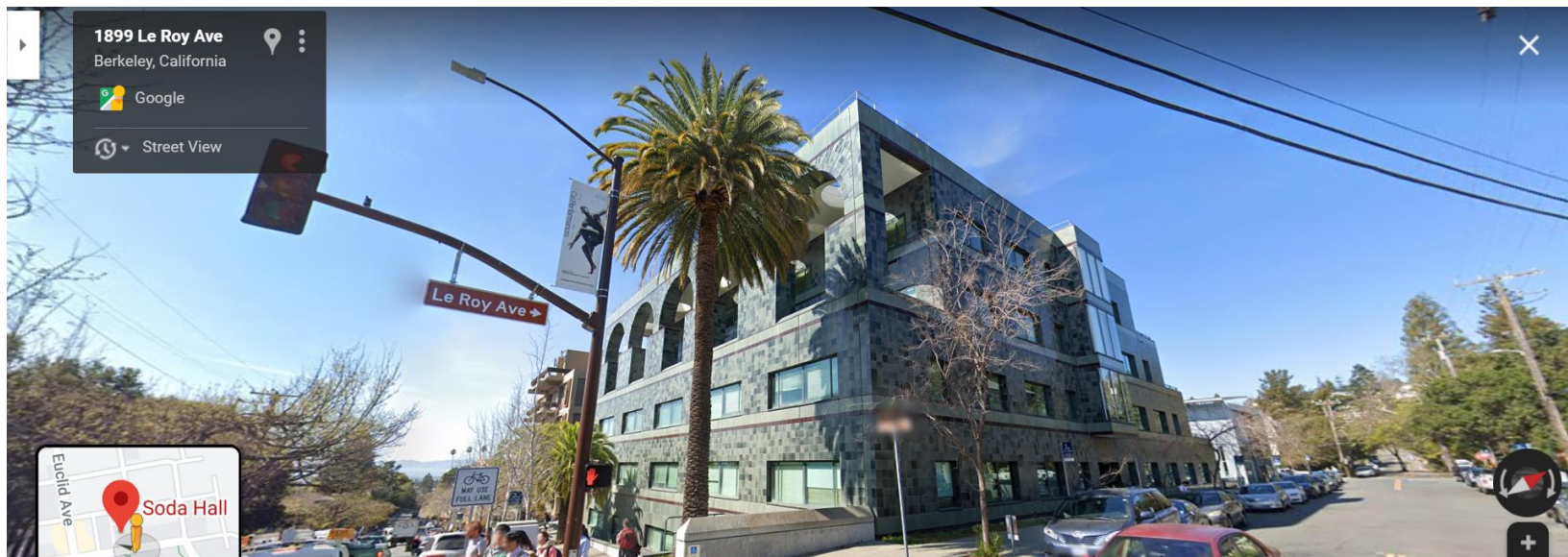
During sunset or
sunrise

After sunset



By Fredo Durand

“Golden Hour”



After sunset: blue hour



Blue Hour (Russian River)

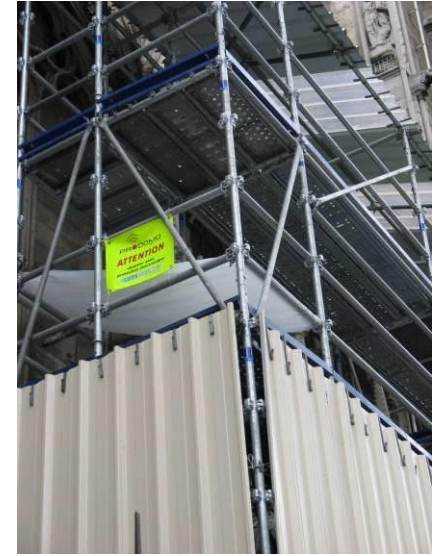
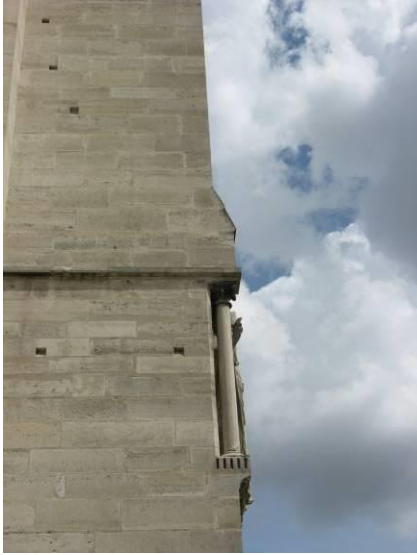


Photo by A. A. Efros

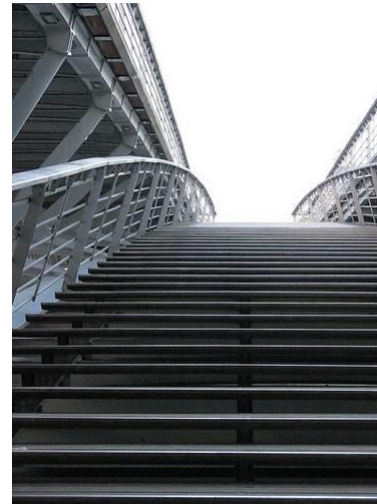
Anyone can take great pictures...



I am a sucky photographer...



...but I am a pretty good photo critic!



<http://flickr.com/photos/aaefros/>

of my Paris photos on Flickr: 32
Total # of my Paris photos: ~1250

~2%

The Postmodern Photographer

The Old Days: a pre-process

- Load film
- Find subject
- Position camera
- Set all the settings “just right”
- Take a deep breath...
- ...Press button!

The New Digital Days: a post-process

- Get a 64 GB memory cartridge
- Take pictures like there is no tomorrow!!!
- ...
- Back home, spend hours of agony trying to find 1-2 good ones

Simplicity to Complexity



Enjoying your next trip to (SF)-MOMA



Pablo Picasso
The Guitar Player (1911)

Simplistic Art PCA

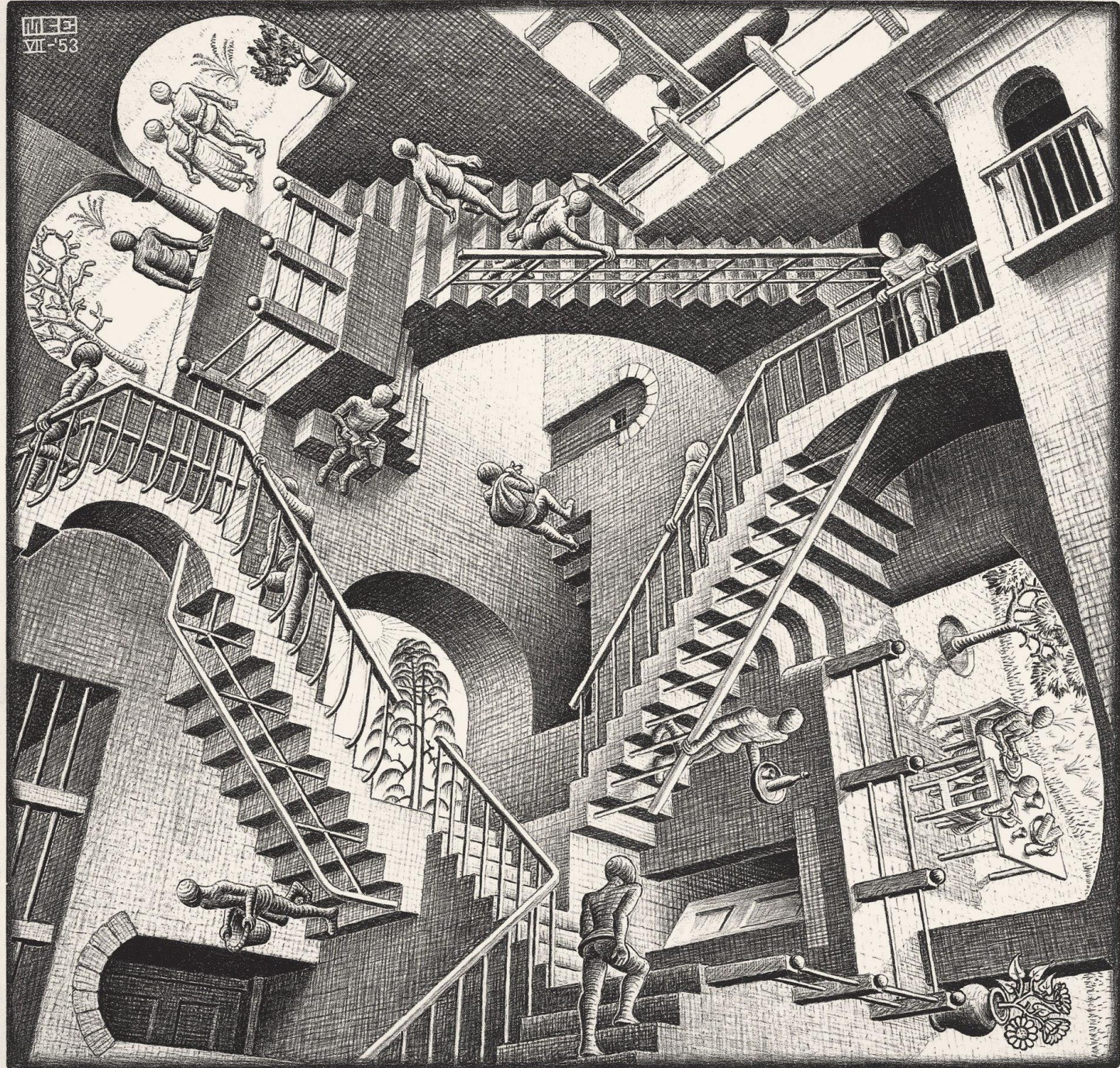
Some “Principal Components” of Art:

- Cute ideas / illusions / clever tricks
- Direct Visceral Pleasure
- Familiarity / Nostalgia
- Art-Historical “Context”

Cute Ideas / Illusions / Clever Tricks

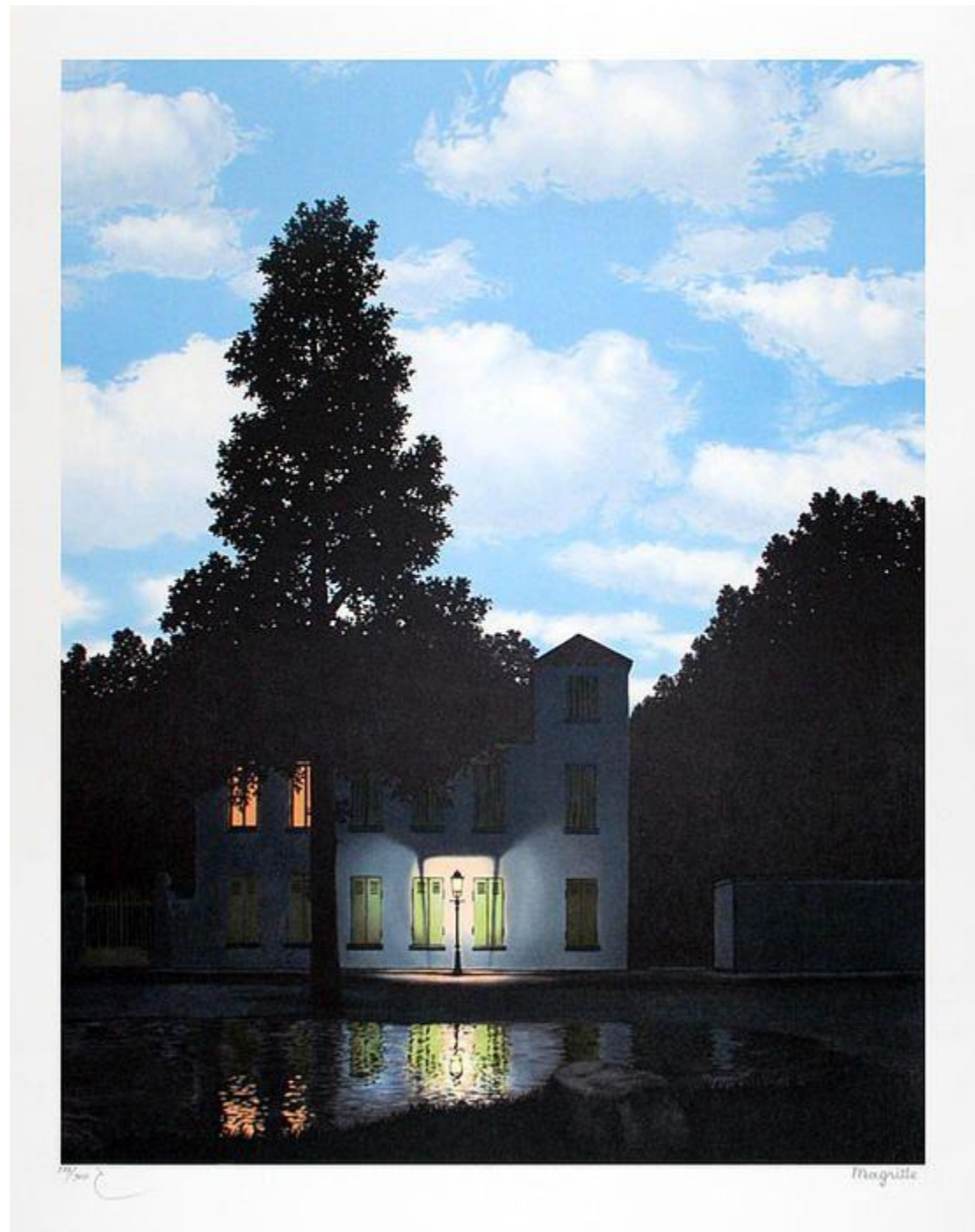


Salvador Dali
*"Gala Contemplating the Mediterranean Sea,
which at 30 meters becomes the portrait
of Abraham Lincoln", 1976*



M.C. Escher

Rene Maritte



People who liked this also liked...



Whangarei Bay
114 km East



Manamotu Islands
2430 km North



Eraser Falls
3514 km North West

Scott Kelly and Ben Polkinghorne

Direct Visceral Pleasure



Jan van Eyck, *The Arnolfini Marriage* (c.1434)



Olafur Eliasson, The Weather Project

Mark Rothko





Mark Rothko

Familiarity / Nostalgia



© Michael Galinsky, 1989
(from "Malls Across America")



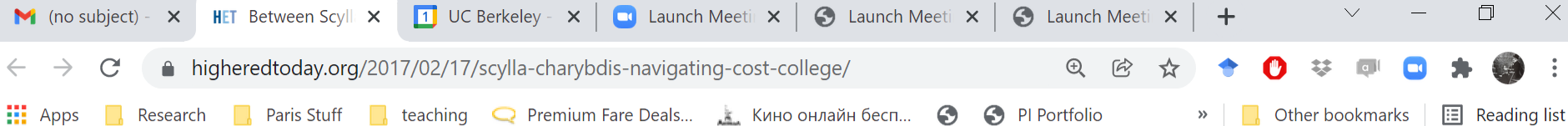
Edward Hopper



Alberto Giacometti

Art-Historical “Context”

Cultural Context



POLICY & RESEARCH ▾

LEADERSHIP ▾

INTERNATIONAL

ATTAINMENT & INNOVATION

Between Scylla and Charybdis: Navigating the Cost of College

February 17, 2017

By Geoffrey L. Brackett

One of my favorite episodes in Homer's *Odyssey* is one of the most fearsome. It involves the sailing between the mysterious cave-dwelling monster Scylla and the terrifying, ship-destroying whirlpool known as Charybdis. There is something about having to manage two supernatural threats at once that makes Odysseus' journey

SHARE THIS
POST

SHARE THIS...



Type here to search



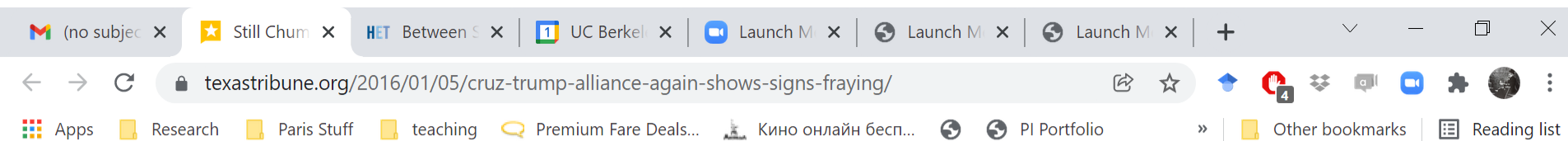
69°F



4:41 PM
12/1/2021



Cultural Context



EXTENDED: Help us unlock \$30,000 in matching funds. Donate now to double your impact.

DONATE

 THE TEXAS TRIBUNE

DONATE

NEWSLETTERS

EVENTS

AUDIO

DATA

LOG IN

SIGN UP



Learning Loss

Abortion Access

Evictions Rising

Find Your Political Districts

COVID-19 Tracker

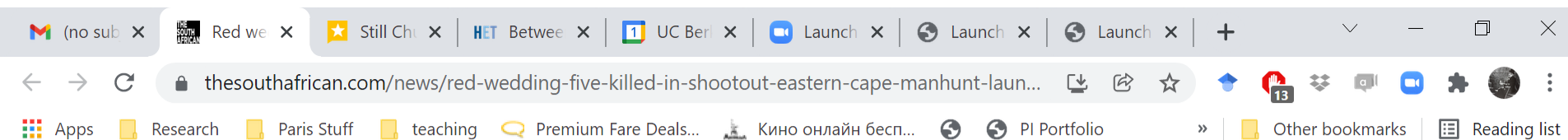
Still Chums? Cruz Says Trump Jumped the Shark

The political bromance between GOP presidential hopefuls Donald Trump and U.S. Sen. Ted Cruz appeared to be unraveling Tuesday after Trump raised the issue of Cruz's Canadian birth, and Cruz suggested his rival has jumped the shark.

BY **PATRICK SVITEK** JAN. 5, 2016 8 PM CENTRAL



Cultural Context



**THE
SOUTH
AFRICAN**

ADVERTISE

ABOUT US

TERMS, CONDITIONS, PRIVACY & GDPR

CONTACT US



NEWS ▾

SPORT ▾

CULTURE ▾

LIFESTYLE ▾

BUSINESS
& FINANCE ▾

#STUDENTLIFE ▾

WIN ▾



Home > News

Red wedding: Five killed in shootout during Eastern Cape nuptials

A manhunt has been launched after gunmen stormed a wedding in the Eastern Cape and opened fire on unsuspecting guests.



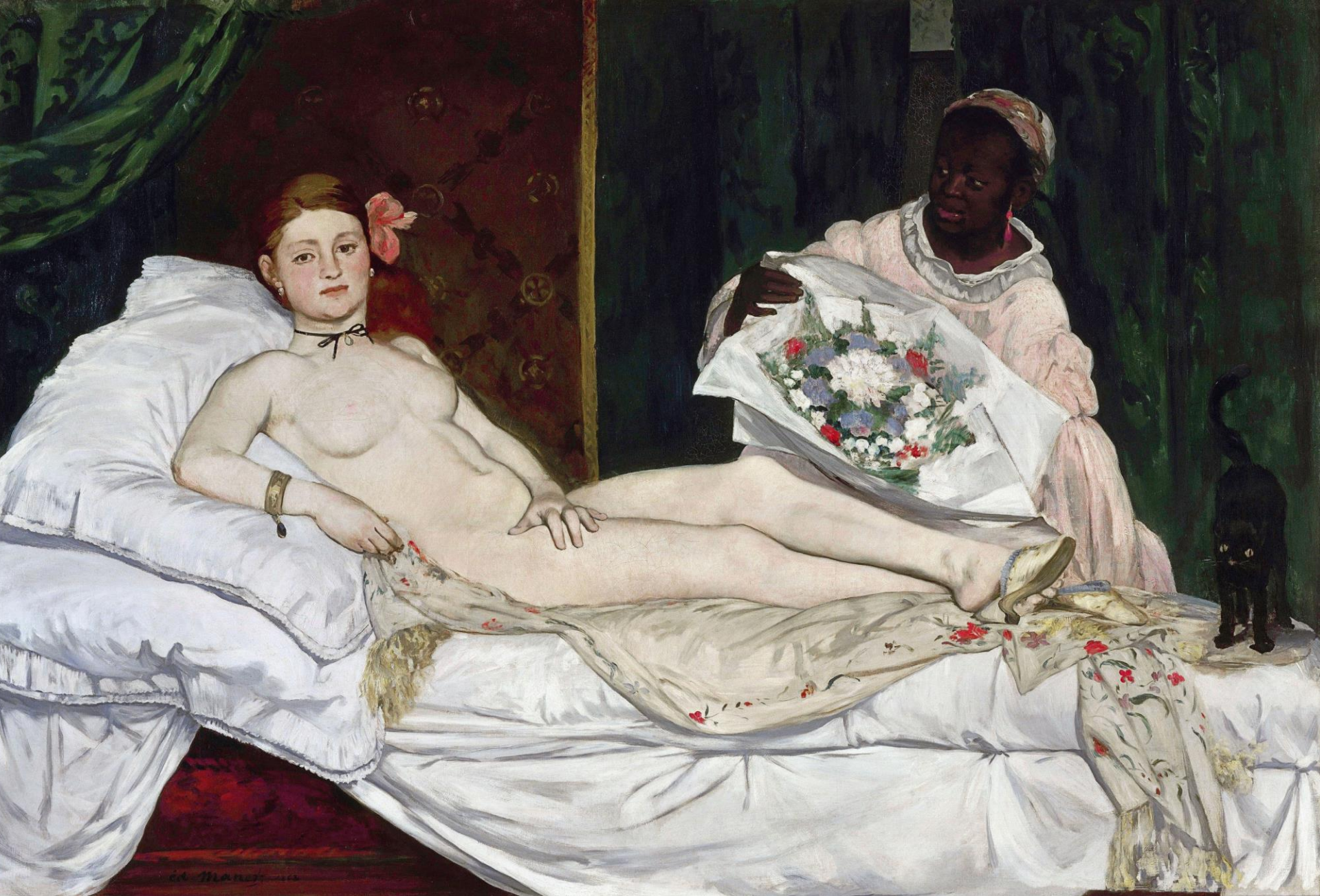
Cultural Context



WIKIPEDIA
The Free Encyclopedia

≡ St. Bartholomew's Day massacre

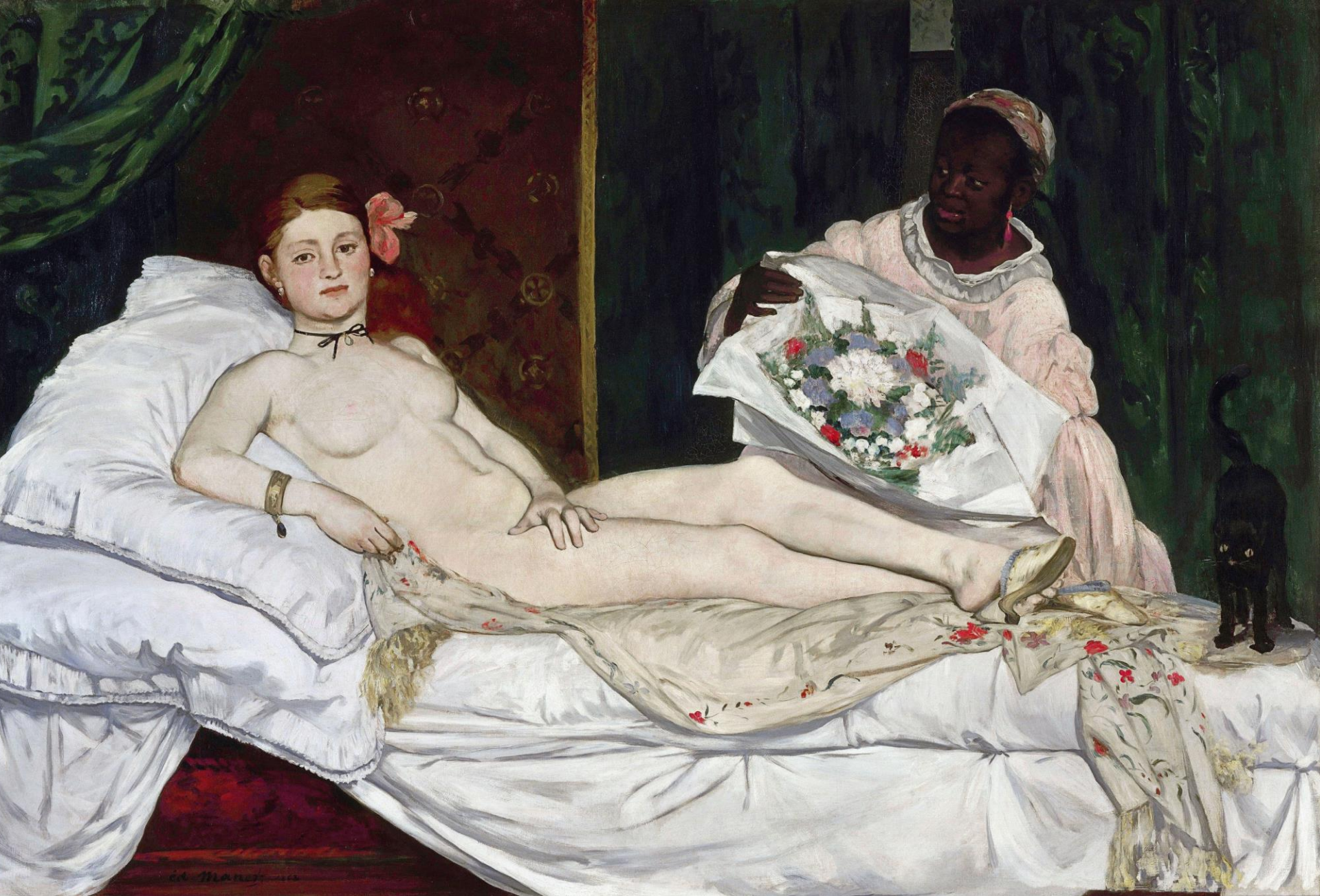
[Article](#) [Talk](#)



Édouard Manet, "Olympia" (1865)



Titian, "Reclining Venus" (1532)



Édouard Manet, "Olympia" (1865)



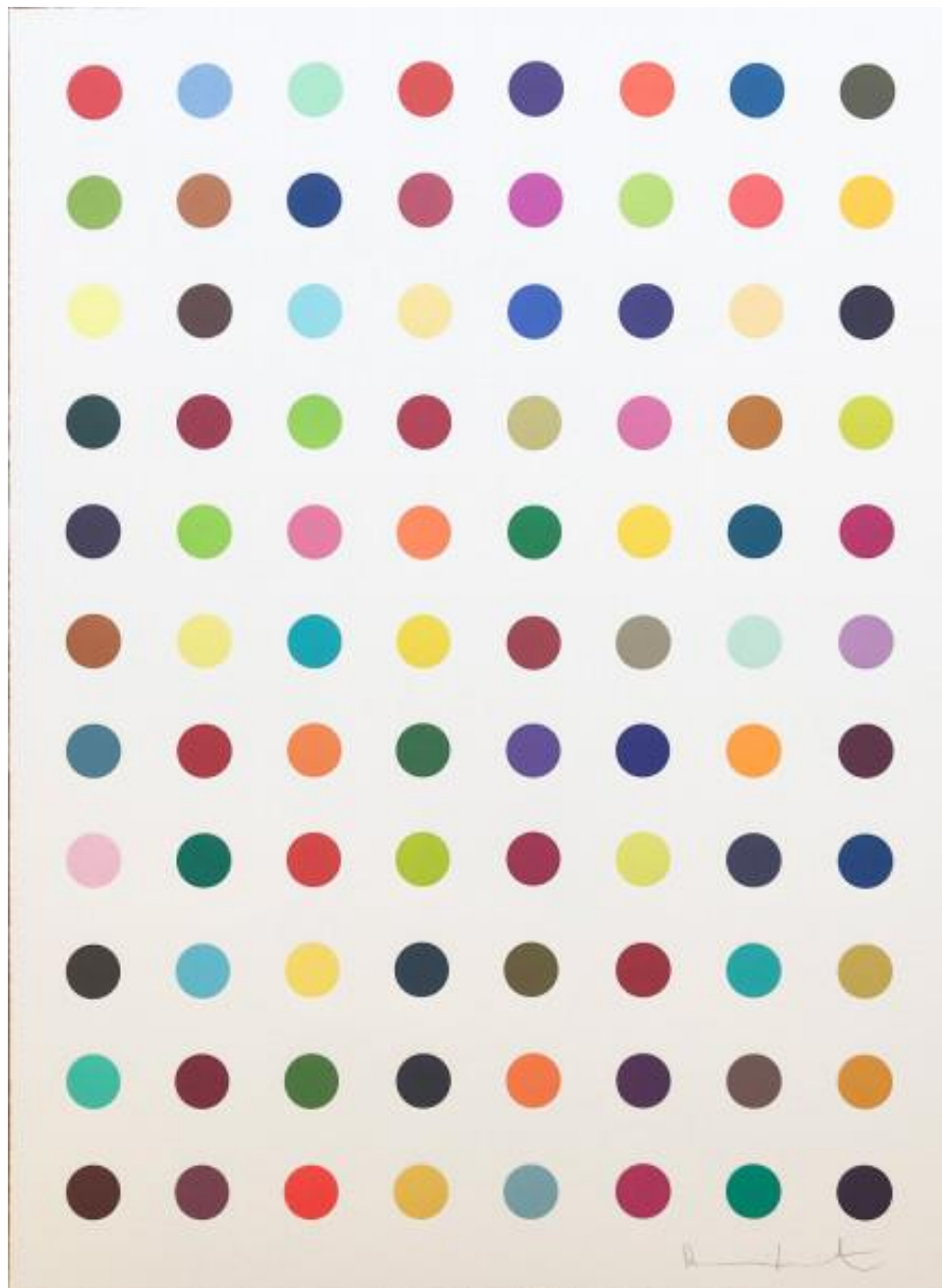
Shepherd Fairey



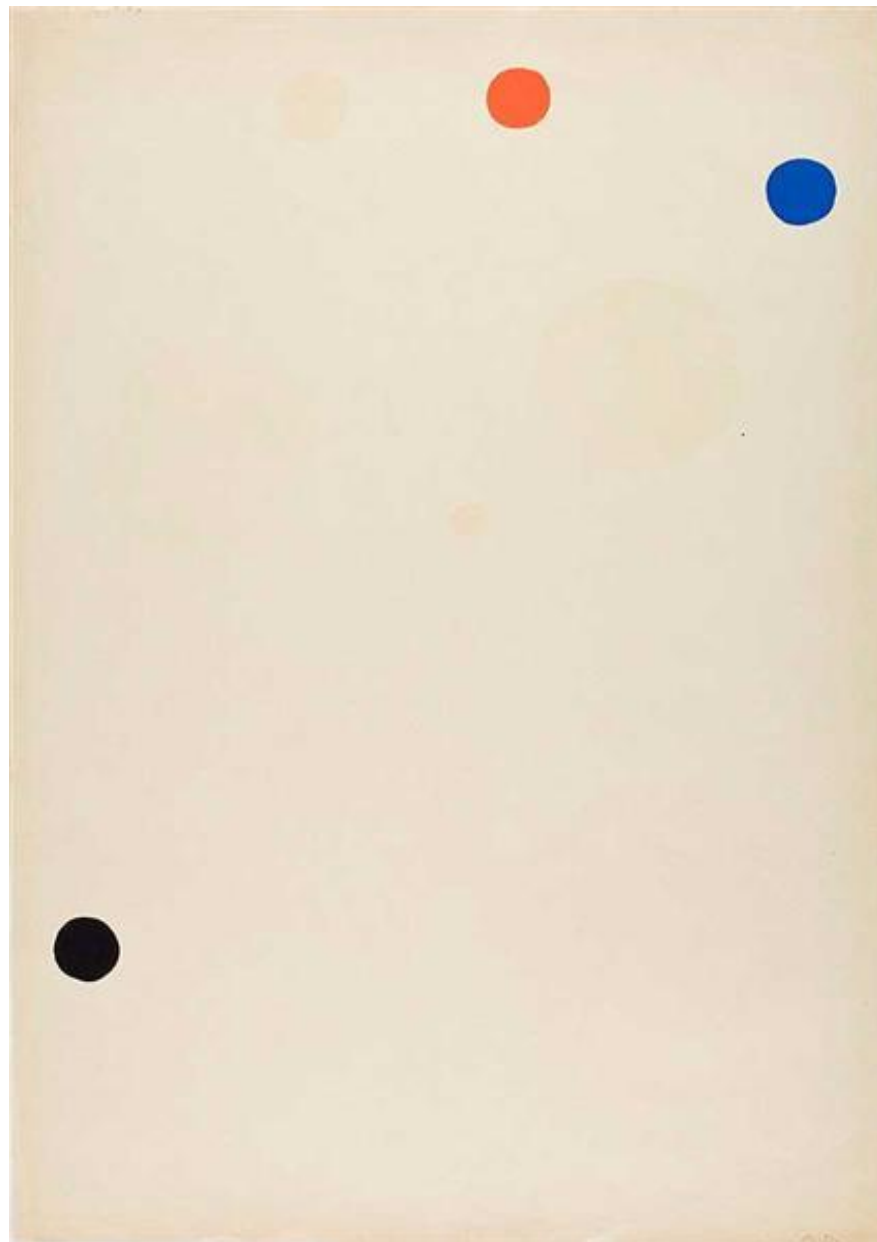
Shepherd Fairey



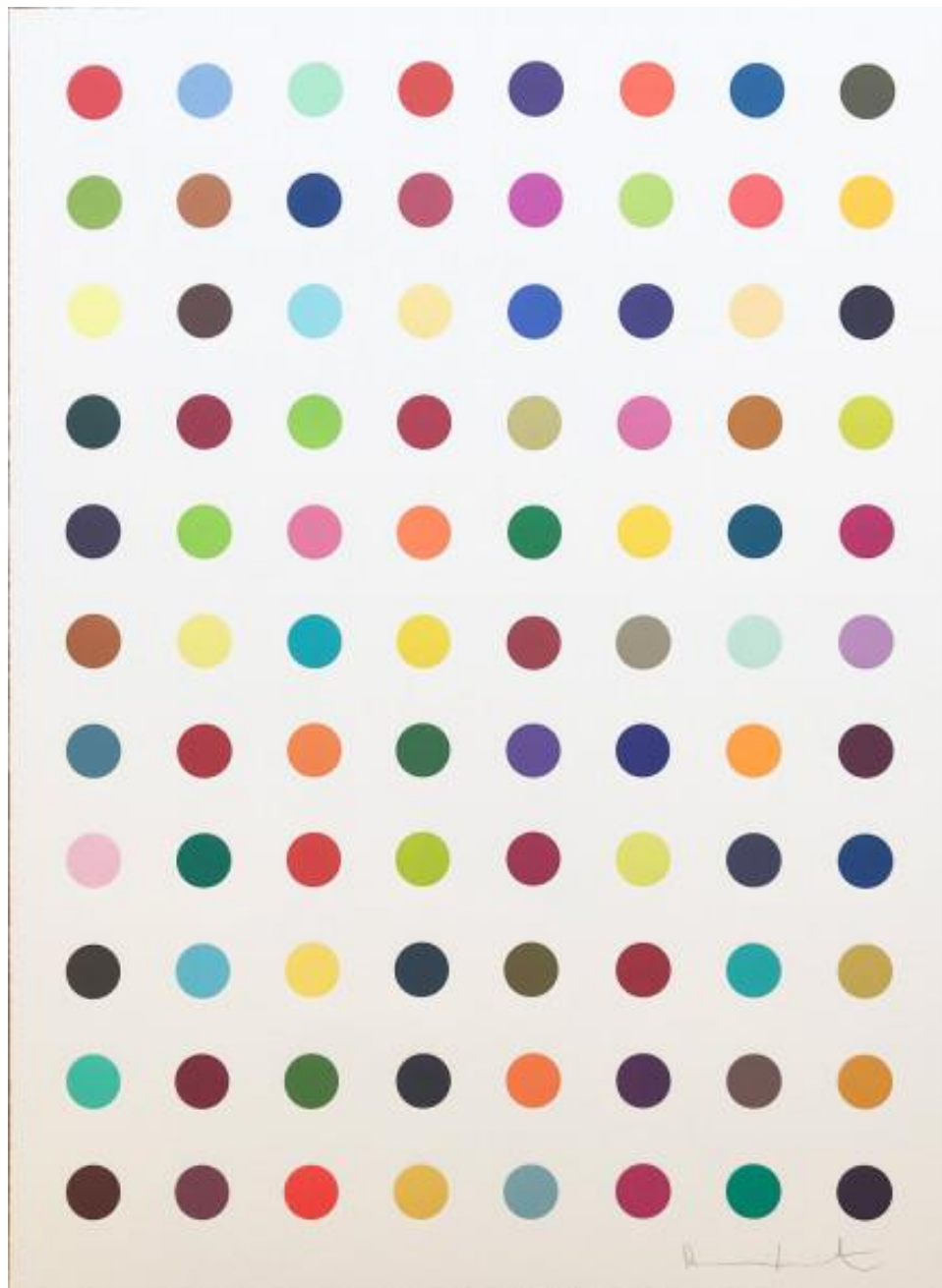
Banksy



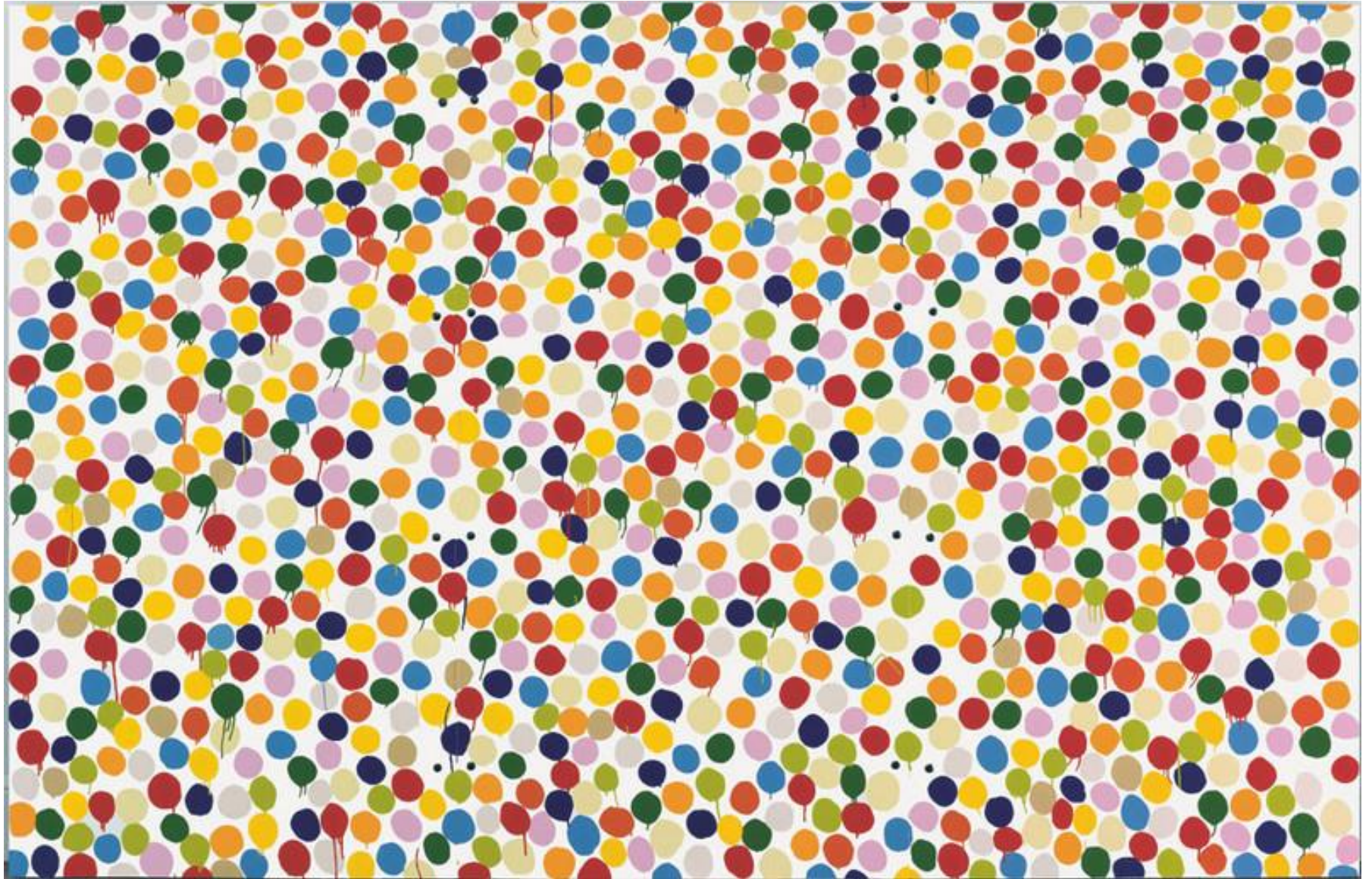
Damien Hirst, Spot Paintings (1968--)



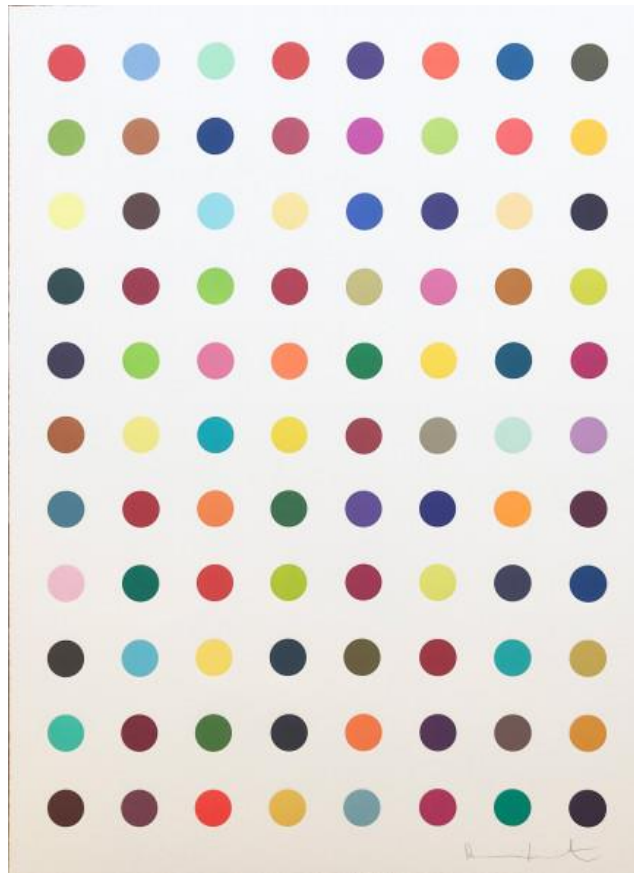
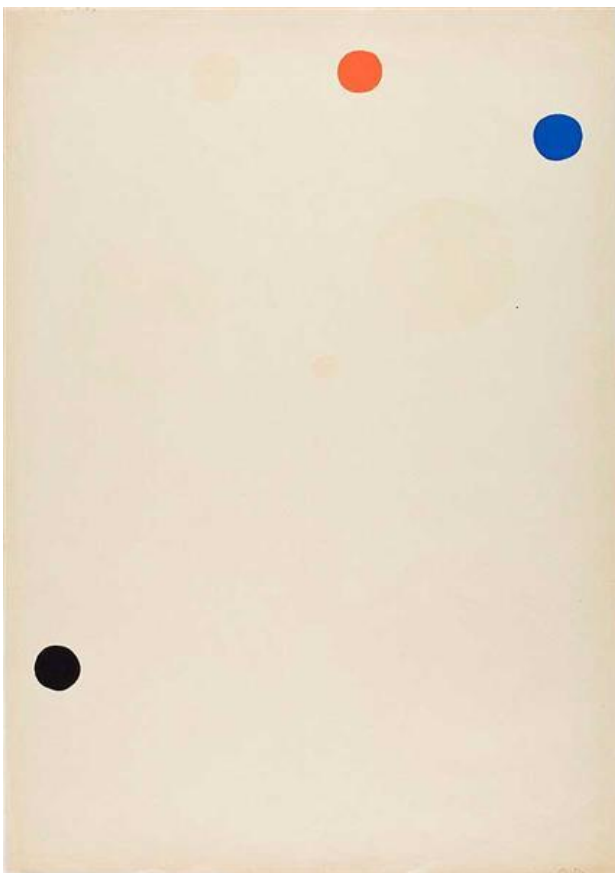
Zlotnikov, "Signals" (1965)



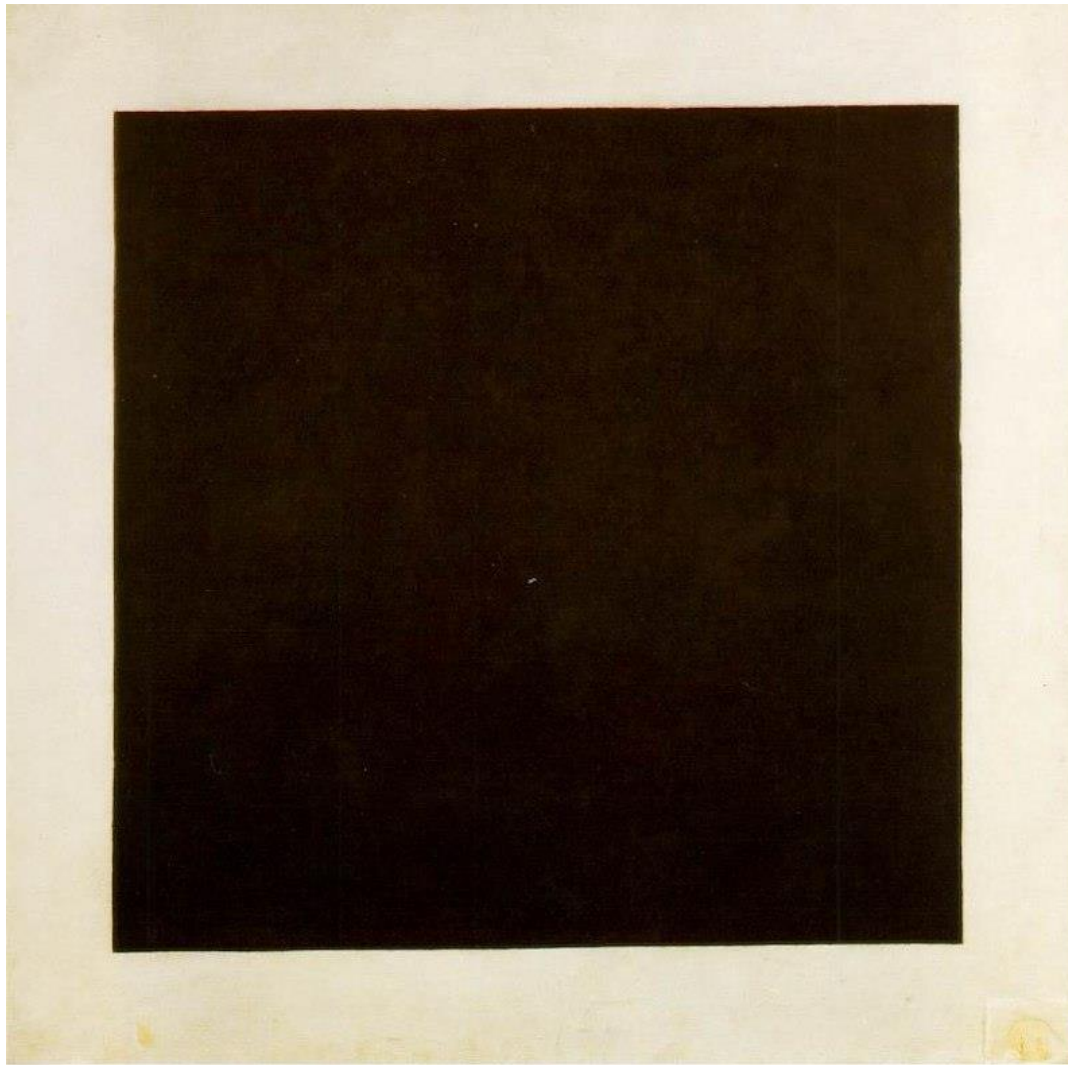
Damien Hirst, Spot Paintings (1968--)



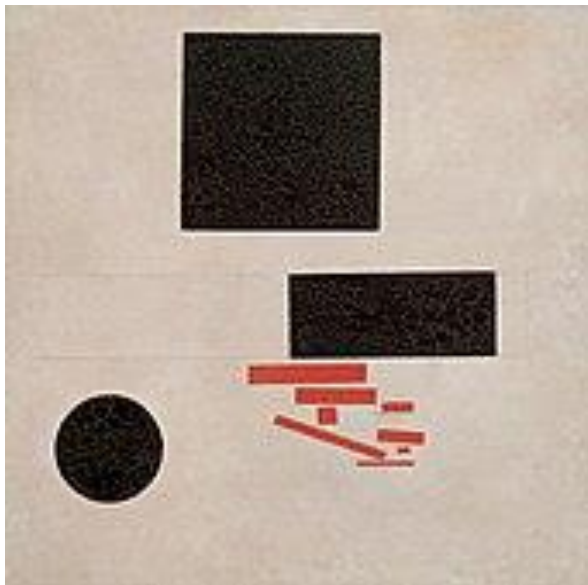
Damien Hirst



Direction of change



Malevich, "Black Square" (1915)



Malevich



“Black Square” as exhibited (1915)

THANK YOU!

